Principal Component Analysis

Principal Component Analysis (PCA)

- PCA is a statistical method used for dimensionality reduction. It transforms a set of possibly correlated variables into a smaller set of uncorrelated variables called principal components, while retaining as much of the dataset's variance (information) as possible.
- It is widely used in machine learning, statistics, signal processing, and data compression.

Purpose of PCA

PCA is used when:

- A dataset has many correlated variables
- We want to reduce dimensionality
- We want to visualize high-dimensional data
- We want to remove redundancy or noise

Example contexts:

- Image compression
- Gene expression analysis
- Finance risk modeling
- Feature extraction before machine learning

Key Idea of PCA

PCA finds new orthogonal axes (directions) that maximize the variance of projected data.

- 1st Principal Component (PC1)

  Direction with maximum variance

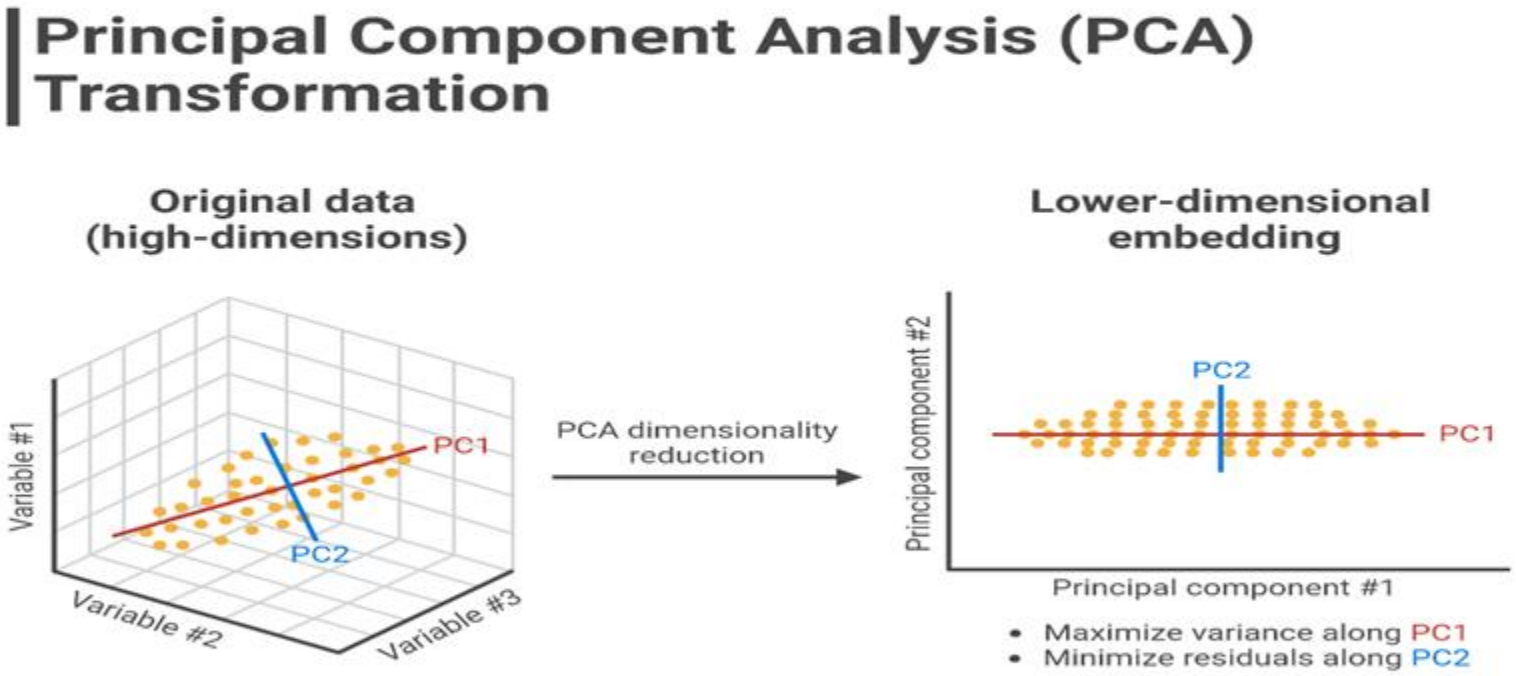
- 2nd Principal Component (PC2)

  Direction orthogonal to PC1 with second highest variance

- And so on.

Mathematically, these directions are given by eigenvectors of the covariance matrix, and the amount of variance captured is given by eigenvalues.

Key Idea of PCA



Mathematical Steps of PCA

Suppose we have dataset X with n observations and p variables.

Step 1 — Standardize Data

Center the data:

$$X_{centered} = X - \bar{X}$$

Where

$\bar{X}$ = mean of each variable.

(Standardization is often done so variables have mean 0 and variance 1.)

Step 2 — Compute Covariance Matrix

$$C = \frac{1}{n-1} X^T X$$

The covariance matrix measures relationships between variables.

Example:

$$C = \begin{bmatrix} Var(x_1) & Cov(x_1, x_2) \\ Cov(x_2, x_1) & Var(x_2) \end{bmatrix}$$

Mathematical Steps of PCA

Step 3 — Eigen Decomposition

We compute **eigenvalues and eigenvectors** of the covariance matrix.

$$Cv = \lambda v$$

Where:

- C = covariance matrix
- v = eigenvector (principal direction)
- λ = eigenvalue (variance explained)

Step 4 — Sort Eigenvalues

Sort eigenvalues in **descending order.**

Largest eigenvalue → **PC1**

Second largest → **PC2**

Mathematical Steps of PCA

Step 5 — Form Projection Matrix

Select top k eigenvectors:

$$W = [v_1 \; v_2 \; \dots \; v_k]$$

Step 6 — Transform Data

Project data into new feature space:

$$Z = XW$$

Z is the **dimension-reduced dataset.**

Mathematical Example with Eigenvalues

Consider a dataset with two variables.

Data matrix:

X	Y
2	0
0	2
3	1
1	3

Step 1: Compute Mean

Mean of X:

$$\bar{x} = \frac{2+0+3+1}{4} = 1.5$$

Mean of Y:

$$\bar{y} = \frac{0+2+1+3}{4} = 1.5$$

Centered data:

X	Y
0.5	-1.5
-1.5	0.5
1.5	-0.5
-0.5	1.5

Mathematical Example with Eigenvalues

Step 2: Covariance Matrix

$$C = \begin{bmatrix} Var(X) & Cov(X,Y) \\ Cov(Y,X) & Var(Y) \end{bmatrix}$$

Compute values:

$$Var(X) = 1.67$$

$$Var(Y) = 1.67$$

$$Cov(X,Y) = -1.33$$

Thus

$$C = \begin{bmatrix} 1.67 & -1.33 \\ -1.33 & 1.67 \end{bmatrix}$$

Variance

$$Var(X) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$Var(Y) = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

Covariance

$$Cov(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Step 3: Eigenvalues

Solve

$$|C - \lambda I| = 0$$

$$\begin{vmatrix} 1.67 - \lambda & -1.33 \\ -1.33 & 1.67 - \lambda \end{vmatrix} = 0$$

$$(1.67 - \lambda)^2 - 1.33^2 = 0$$

Solutions:

$$\lambda_1 = 3$$

$$\lambda_2 = 0.34$$

Interpretation:

- PC1 explains **most variance (3)**
- PC2 explains **little variance (0.34)**

Mathematical Example with Eigenvalues

Step 4 — Use Eigenvalue Equation

Eigenvectors satisfy

$$(C - \lambda I)v = 0$$

Substitute $\lambda_1 = 3$.

$$C - 3I = \begin{bmatrix} 1.67 - 3 & -1.33 \\ -1.33 & 1.67 - 3 \end{bmatrix}$$

$$= \begin{bmatrix} -1.33 & -1.33 \\ -1.33 & -1.33 \end{bmatrix}$$

Step 5 — Solve the Linear System

Let the eigenvector be

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

Now solve

$$(C - 3I)v = 0$$

$$\begin{bmatrix} -1.33 & -1.33 \\ -1.33 & -1.33 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Multiplying:

$$-1.33x - 1.33y = 0$$

Divide by −1.33:

$$x + y = 0$$

Thus

$$y = -x$$

Mathematical Example with Eigenvalues

Step 6 — Choose a Convenient Vector

Any vector satisfying

$$y = -x$$

is an eigenvector.

Example choice:

$$v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 7 — Normalize the Eigenvector

In PCA we usually normalize eigenvectors to unit length.

Length:

$$\begin{aligned} ||v|| &= \sqrt{1^2 + (-1)^2} \\ &= \sqrt{2} \end{aligned}$$

Normalized vector:

$$\begin{aligned} v_1 &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix} \end{aligned}$$

Approximate:

$$v_1 \approx \begin{bmatrix} 0.71 \\ -0.71 \end{bmatrix}$$

Mathematical Example with Eigenvalues

Step 8: Eigenvectors

For $\lambda_1 = 3$

Eigenvector approximately

$$v_1 = \begin{bmatrix} 0.71 \\ -0.71 \end{bmatrix}$$

For $\lambda_2 = 0.34$

$$v_2 = \begin{bmatrix} 0.71 \\ 0.71 \end{bmatrix}$$

Step 9: Principal Components

The new coordinate axes are:

PC1:

$$z_1 = 0.71x - 0.71y$$

PC2:

$$z_2 = 0.71x + 0.71y$$

Most of the variance lies along **PC1**.

Thus we can reduce from **2 dimensions → 1 dimension** by keeping only **PC1**.

Mathematical Example with Eigenvalues

Step 10: Variance Explained

Total variance:

$$\lambda_1 + \lambda_2$$

Variance explained by PC1:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$= \frac{3}{3 + 0.34}$$

$$= \frac{3}{3.34}$$

$$= 89.8\%$$

Thus **one component keeps ~90% of the information.**

Geometric Interpretation

PCA finds a new rotated coordinate system where:

- Axis 1 = maximum spread
- Axis 2 = orthogonal residual spread

Visually:

- Data cloud is rotated
- Projection onto main axis preserves most information

Advantages of PCA

- Reduces dimensionality
- Removes multicollinearity
- Improves computational efficiency
- Helps visualization
- Noise reduction

Limitations

- Linear method only
- Components may be hard to interpret
- Sensitive to scaling
- Sensitive to outliers

Applications

- Machine Learning
  Feature extraction before algorithms.
- Computer Vision
  Face recognition (Eigenfaces).
- Finance
  Portfolio risk modeling.
- Bioinformatics
  Gene expression analysis.
- Image Compression
  Reducing image dimensions.

Thank You